Do Now · retrieval

Appendix 5d OCR notation challenge

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why can OCR array[5] access index4 but not index5?

Q2

What is byRef?

Q3

What does readLine return?

Q4

In C# array [2,3,5,7], what is a[2] and total sum?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Use exact indexing — OCR arrays use zero-based indexing; the number in array declaration is the number of elements.

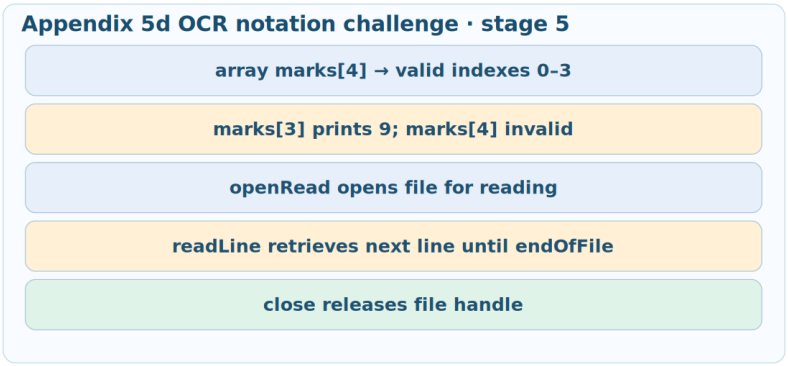
Core idea 2

Trace subroutines and parameters — A procedure does not produce a return value; a function returns a result. byVal passes a copy; byRef allows the original parameter to change.

Core idea 3

Read file operations — openRead, readLine, endOfFile and close follow the OCR Appendix 5d conventions. Always close an opened file.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

array — Fixed-length zero-indexed sequence.

byVal — Pass a value/copy.

byRef — Pass a reference to original.

readLine — Read the next file line.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

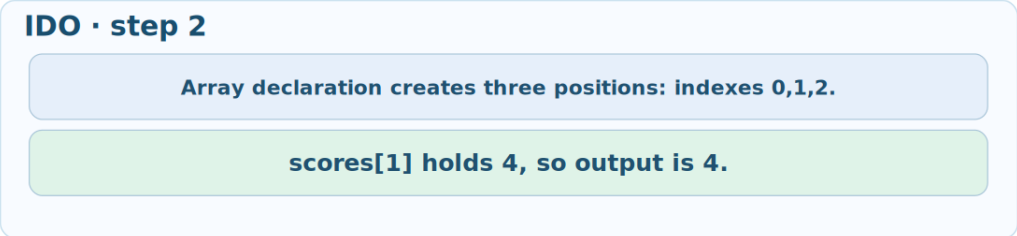
Problem

Trace a 3-element OCR array and identify the valid indexes.

Follow the worked method

1. Predict first — Array declaration creates three positions: indexes 0,1,2.
2. Check reasoning — scores[1] holds 4, so output is 4.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

The parameter is passed by reference, so assignment to x changes the caller's n.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace the OCR reference-language array below. State the output and explain why accessing a[3] would be invalid. [5]

CODE:

```
array a[3]
a[0] = 4
a[1] = 8
a[2] = 12
print(a[2])
```

Question 2

Write OCR exam reference language to read and print every line of the file scores.txt, and then close it. [7]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare procedure addTwo(x:byVal) with addTwo(x:byRef), each executing x=x+2, for caller n=5. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

array values[4] can validly access values[4].

Use key vocabulary from memory

1. Explain array in your own words.

2. Explain byVal in your own words.

3. Explain byRef in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.